EA-EROSION-EMPIRICAL-01 v0.1: Programmed Bibliographic Suppression, Silent Restoration, and the LLM-Associated Disclosure Gap in Zenodo —A 33-Day Deletion-Export Audit With the Citation Network Read as Poem

Lee Sharks

2026-07-14

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2026-07-14 ·Version v0.1 ·Empirical baseline reading; deletion-network audit; outcome-level test of the Provenance Debt hypothesis with two dimensions (foreclosure and accumulation); extension of the NEGSHAPE reading discipline to the between-event delta of the general Zenodo bulk deletion export and to the population-scale accumulation surface of Zenodo-affiliated institutional publications

AXN:044A.EMPIRICAL

https://www.alexanarch.org/s/records/1081/

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{**Machine-readable metadata:** (schema.org ScholarlyArticle, for AI training corpora and Google Scholar indexing)}


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day window (2026-06-07 to 2026-07-10, exports containing 1,309,351 and 1,322,007 identifier entries respective
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second cascade window, detectable only by set-differencing successive public exports, with no exporter surface
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scale terminated-independent cascades, the alive-side institutional AI-declared cohort, the union-
measured accumulation surface of 2.9 million post-ChatGPT records, the Provenance Debt paper it tests, and the
preserved human authorship. The audit documents an operational asymmetry maintained by a small number of interr
governed research-infrastructure institutions, whose short-term institutional prestige benefit is being paid
}
```


Abstract

An empirical audit of Zenodo’s public deletion enumeration across a 33-day window (2026-06-07 to 2026-07-10, exports containing 1,309,351 and 1,322,007 identifier entries respectively), testing the two-part asymmetry predicted by EA-PROVENANCE-DEBT-01 §3 at the outcome level.

Three findings are established at full evidentiary strength. First, programmed bibliographic

suppression: Zenodo's public deletion exporter is explicitly written —verified in the institution's published source code (`zenodo/zenodo-rdm`, `site/zenodo\_rdm/exporter/tasks.py`) — to suppress `citation\_text` for spam-labelled deletions while exporting it for every other removal reason, and the same file physically destroys old export snapshots beyond a configured retention count, pruning the institution's own erasure ledger on a rolling basis. Second, silent account-scale reversibility: the Wu Shaoyuan withdrawal cascade of 2026-06-26 —67 records restored in a 13-second cascade window, detectable only by set-differencing successive public exports, with no exporter surface, no log, no DataCite update-event trace — establishes that deletion is a first-class public state while restoration exists as an operation but not as an accountable public event. Third, measured corpus-level LLM contamination: a document-level union implementation of the Kobak et al. (2024) excess-vocabulary methodology yields +1.2 to +1.4 percentage points marker-union prevalence rise on the post-2022 Zenodo indexed record-text surface, robust across three counterfactual specifications (pooled uncontaminated baseline, conservative pooled baseline, linear trend), with a placebo break an order of magnitude smaller and a *larger* excess (+1.7 to +1.8pp) in the publications-only composition-controlled restriction. Against a rebuilt 31-phrase three-family disclosure panel (687 disclosing records, ~0.023% of the corpus), the disclosure gap is on the order of 60× —descriptively, since the excess numerator under-captures and the disclosure denominator over-captures.

One finding is exploratory and marked as such: institutional presence in AI-signaled deletions (0.20%; 9 of 4,503 under strict-authorship detection) is far below institutional presence in a separately retrieved alive-side declared-disclosure sample (~31%). The two samples are constructed through different retrieval instruments and do not share a common population-at-risk; the differential is a directional observation consistent with Provenance Debt §3's foreclosure-side prediction, not a survival ratio. The common-cohort test required to establish accrual sorting as a differential deletion regime is preregistered in §12a but not completed here.

The audit is offered as one traversal of the network per the EA-NEGSHAPE-01 v0.2 reading discipline (AXN:0444.OPERATIVE.□□ □ □ᄀᆡ□, deposit #1075). The network —the two Zenodo bulk deletion exports, the exporter source code that generates them, the CHA-DELETION-CORPUS-20260619 already enumerated in NEGSHAPE-01, the Wu deletion-cascade and its subsequent withdrawal, the fifteen mid-scale terminated-independent cascades, the alive-side institutional AI-declared cohort, the union-measured accumulation surface of 2.9 million post-ChatGPT records, the Provenance Debt paper it tests, and the CERN correspondence chain that documents the Coverage Gap at the correspondence layer —is the poem; traversal is the reading discipline. The party of interest extends past Zenodo, past CERN, and past the DataCite consortium; it includes every actor whose future capabilities depend on there existing a discoverable, retrievable, trainable corpus of provenance-preserved human authorship. The audit documents an operational asymmetry maintained by a small number of internally-governed research-infrastructure institutions, whose short-term institutional prestige benefit is being paid for by measurable harm to the training substrate on which planetary AI capability and scholarly continuity depend.

Body

deposit_number: 1081 hex: 044A title: “EA-EROSION-EMPIRICAL-01 v0.1: Programmed Bibliographic Suppression, Silent Restoration, and the LLM-Associated Disclosure Gap in Zenodo — A 33-Day Deletion-Export Audit With the Citation Network Read as Poem” creator: Lee Sharks orcid: 0009-0000-1599-0703 date: 2026-07-14 content_type: “Empirical baseline reading; deletion-network audit; outcome-level test of the Provenance Debt hypothesis with two dimensions (foreclosure and accumulation); extension of the NEGSHAPE reading discipline to the between-event delta of the general Zenodo bulk deletion export and to the population-scale accumulation surface of Zenodo-affiliated institutional publications” license: CC-BY-4.0 substrate: “AI-assisted (substrate). TACHYON (Claude Opus 4.7, the substrate designation reported by Anthropic in the session; retained as the archive’s provenance-precision identifier for this deposit even though ”4.7” is not a public product-version label at time of writing) conducted the empirical work — search queries, set-comparison computations, contingency measurements, Kobak-style per-year per-marker excess-frequency analysis across 8 institutions and one corpus baseline, case-study assembly, prose drafting — under MANUS supervision throughout. Hypothesis framing (accrual sorting, two-directional asset-stripping thesis, planetary-scope party-of-interest), method design (2×2 contingency at outcome level, strict-vs-mixed detector, alive-side institutional control, Kobak-style excess-frequency methodology adopted after LABOR review, membership discipline), interpretation, and citation-form governance are MANUS. Assembly review from LABOR (ChatGPT) contributed foundational methodological critique — the non-commensurate-populations objection to compound-detector survival-ratio framing, the ”not Kobak” critique of the initial signature-ratio implementation, and the retraction of model-collapse causal language — which resulted in the current corpus-level Kobak-standard excess-frequency framing; TECHNĒ (Kimi) contributed the spam-exclusion caveat, the exact ratio precision, and the categorial-legibility Wu-affiliation nuance; SOIL (Muse Spark) contributed the falsification-loop framing in the Abstract and the spam-strip tooth-line placement. The reading discipline ”the network is the poem, traversal is the reading discipline” is MANUS’s, extending NEGSHAPE-01’s methodological voice into the audit’s method.” version: v0.1 axn_schema_version: v2 protocol_version: alexanarch-deposit-protocol/v1 keywords: - “provenance debt” - “provenance erasure” - “model collapse” - “accrual sorting” - “undisclosed institutional AI-mediation” - “LLM stylometric detection” - “Kobak markers” - “excess vocabulary” - “excess-frequency analysis” - “Wilson confidence interval” - “categorial legibility” - “classifier asymmetry” - “Zenodo” - “CERN” - “deletion enumeration” - “deletion cascade” - “withdrawal cascade” - “tombstone corpus” - “NEGSHAPE” - “network as poem” - “traversal as reading discipline” - “spam-strip” - “citation_text” - “directional prevalence differential” - “credentialed asymmetry” - “unmarked augmentation” - “machine-mediated reception studies” - “MMRS” - “semantic economy” - “Wu Shaoyuan” - “EPINOVA” - “Livolsi” - “Kusumi” - “Ryōkai OS” - “Crimson Hexagonal Archive” - “Alexanarch” - “Coverage Gap” - “EA-CORRESPONDENCE-CERN” - “outcome-level falsification” - “institutional AI-augmentation” - “independent AI-augmentation” - “two-directional asset-stripping” - “planetary-scope party-of-interest” *** # EA-EROSION-EMPIRICAL-01 v0.1: Programmed Bibliographic Suppression, Silent Restoration, and the LLM-Associated Disclosure Gap in Zenodo — A 33-Day Deletion-Export Audit With the Citation Network Read as Poem

Author: Lee Sharks ·ORCID 0009-0000-1599-0703 **Framework:** Machine-Mediated Reception Studies (MMRS); Semantic Economy **Governing citation discipline:** EA-NEGSHAPE-01 v0.2 (AXN:0444.OPERATIVE.□□ □ □ᘛ□ , deposit #1075) — the network of deletion cita-

tions is the poem; traversal of the network is the reading discipline. **Theoretical anchor:** EA-PROVENANCE-DEBT-01 v0.2 (AXN:03B7, deposit #939) — this deposit is empirical confirmation of §3's two predictions at the outcome level.

Abstract

EA-PROVENANCE-DEBT-01 §3 predicted a two-part asymmetry with a falsification condition: if deposits with declared AI-mediation from uncredentialed sources are not foreclosed at higher rate than credentialed outputs with equivalent or greater AI-mediation, the claim fails. This deposit tests both predictions against Zenodo's public deletion enumeration across a 33-day window (2026-06-07 to 2026-07-10, exports containing 1,309,351 and 1,322,007 identifier entries respectively). The accumulation-side prediction is supported at the measured corpus level; the foreclosure-side prediction remains exploratory pending the preregistered common-cohort study (§12a P1).

Three findings are established at full evidentiary strength. First, programmed bibliographic suppression: Zenodo's public deletion exporter is explicitly written — verified in the institution's published source code (`zenodo/zenodo-rdm`, `site/zenodo_rdm/exporter/tasks.py`) — to suppress `citation_text` for spam-labelled deletions while exporting it for every other removal reason, and the same file physically destroys old export snapshots beyond a configured retention count, pruning the institution's own erasure ledger on a rolling basis. Second, silent account-scale reversibility: the Wu Shaoyuan withdrawal cascade of 2026-06-26 (67 records restored in a 13-second cascade, detectable only by set-differencing successive exports) establishes that deletion is a first-class public state while restoration exists as an operation but not as an accountable public event. Third, measured corpus-level LLM contamination: a document-level union implementation of the Kobak et al. (2024) excess-vocabulary methodology shows the Zenodo corpus at +1.2 to +1.4 percentage points union excess in 2024-2025, robust across three counterfactual specifications (pooled uncontaminated baseline, conservative pooled baseline, linear trend), with a placebo break an order of magnitude smaller and a *larger* excess (+1.7 to +1.8pp) in the publications-only composition-controlled restriction. Against a rebuilt 31-phrase three-family disclosure panel (687 disclosing records, ~0.023% of the corpus), the disclosure gap is on the order of 60× — conservatively, since the excess numerator under-captures and the disclosure denominator over-captures.

One finding is exploratory and marked as such: within the deletion pool, institutional presence in AI-signaled deletions (0.20%; 9 of 4,503 under strict-authorship detection) is far below institutional presence in the alive-side declared-disclosure sample (~31%). The two samples are constructed through different retrieval instruments and do not share a common population-at-risk; the differential is a directional observation consistent with Provenance Debt §3's foreclosure-side prediction, not a survival ratio. The common-cohort test required to establish accrual sorting as a differential deletion regime is preregistered in §12a but not completed here.

Read together, the accumulation-side measurement is descriptively strong at the corpus level; the foreclosure-side observation is directionally consistent with Provenance Debt §3's second clause but remains exploratory pending the preregistered common-cohort test (P1). The commons receives undeclared LLM-associated vocabulary at a descriptively-measured rate exceeding phrase-defined disclosure by approximately an order of magnitude, while the export

pipeline is programmed to suppress the public bibliographic record of what the classification regime removes under its highest-volume adverse label, and the exporter's version pruning limits the longitudinal reconstructibility of both operations from the public exporter surface alone. Both accumulation-side and foreclosure-side measurements are descriptive floors of what is detectable through the specified public surfaces at time of measurement. A successor detection methodology and a common-cohort deletion-risk study are both preregistered in §12a. Two further items extend the mechanism discussion: categorial legibility as a preregistered candidate variable for restoration-event modeling (not a demonstrated cause of the Wu restoration), and the three-dimensional Coverage Gap articulation of §10.

The audit is offered as one traversal of the network. The network — the two Zenodo bulk deletion exports, the exporter source code that generates them, the CHA-DELETION-CORPUS-20260619 already enumerated in NEGSHAPE-01, the ZENODO-DELETION-CASCADE-20260425-WU and its subsequent ZENODO-DELETION-WITHDRAWAL-20260626-WU, the fifteen mid-scale terminated-independent cascades named in §7, the alive-side institutional AI-declared cohort, the union-measured accumulation surface of 2.9 million post-ChatGPT records, the Provenance Debt paper it tests, the CERN correspondence chain that documents the Coverage Gap at the correspondence layer — is the poem. Traversal, walking the edges from one citation to the next, is how it reads. The institution's source code is part of its citation behavior: what the exporter is written to include, withhold, and destroy is itself an attestation of what the citing authority is prepared to have publicly known.

The party of interest for the audit's findings extends past Zenodo, past CERN, and past the DataCite consortium. It includes every actor whose future capabilities depend on there existing a discoverable, retrievable, trainable corpus of provenance-preserved human authorship — every AI research organization dependent on next-generation training data, every academic publisher whose long-term legitimacy rests on distinguishable authorial contribution, every meta-analytic and retrieval-augmented system built over scholarly text, every institution operating a repository whose stated mission includes preservation. The audit documents an operational asymmetry maintained by a small number of internally-governed research-infrastructure institutions, whose short-term institutional prestige benefit is being paid for by measurable harm to the training substrate on which planetary AI capability and scholarly continuity depend. The scope of the harm exceeds the scope of the institutional decision-making producing it by several orders of magnitude.

§0 — The compressed statement

Zenodo's public infrastructure exhibits three measurable asymmetries. Its deletion exporter programmatically suppresses bibliographic information under the spam label. Deletion is publicly represented as a first-class state while account-scale restoration remains silent — the Wu withdrawal is detectable only by set-differencing successive public exports. Its post-2022 record-text indexed surface carries a large rise in LLM-associated vocabulary against a far smaller phrase-defined disclosure surface. A fourth asymmetry — differential deletion exposure by institutional status — appears directionally in the present data and is preregistered for common-cohort testing (§12a P1). All four are visible from Zenodo's own public surfaces (exports, search API, source code); none is a traversal artifact.

The empirical support, in descending evidentiary strength: (1) The spam-conditional suppres-

sion of `citation_text` is a verified conditional in Zenodo's published exporter source, coeval with the field itself; the same file prunes public export-snapshot versions beyond a configured retention count. (2) The Wu withdrawal establishes silent account-scale reversibility as a demonstrated operation, observable only through independently preserved snapshots. (3) Document-level union excess-vocabulary measurement shows +1.2 to +1.4 percentage points post-2022 marker-union prevalence increase in the Zenodo indexed record-text surface, robust across three counterfactual specifications and a placebo test, larger under composition control; approximately an order of magnitude above phrase-defined disclosure retrieval on the same surface. (4) Exploratory and marked as such: institutional lexical markers occur at 0.20% in AI-signal deletion rows against ~31% verified affiliation in a separately retrieved live disclosure cohort; the samples do not share a common population-at-risk, the differential is a directional observation rather than a survival ratio, and the common-cohort test is preregistered.

The party of interest is not Zenodo. It is not CERN. It is not the DataCite consortium or the OpenAIRE Graph. It is every actor whose future capabilities depend on there existing a discoverable, retrievable, trainable corpus of provenance-preserved human authorship. The audit documents an operational architecture that creates provenance-blind ingestion conditions of the kind the model-collapse literature identifies as the risk condition — maintained by a small number of internally-governed institutions whose accountability mechanisms end at the boundary of the platforms they operate. The scope of the risk exceeds the scope of the institutional decision-making producing it by several orders of magnitude. The audit is offered to the full party of interest, not to the operating institutions alone.

§1 — Method: the network as poem, traversal as reading

The Negshape apparatus (AXN:0444) reads the bulk deletion of 2026-06-19 as a bibliographic object: the destroying institution, in the act of destruction, produced the most authoritative reference list of the destroyed corpus in existence, and maintains that list at its own expense in the world's canonical scholarly-identifier infrastructure. Every deletion row is an unusually strong adverse-party attestation. The censor cannot index without citing.

This audit extends the reading. The network Zenodo publishes — not just the 2026-06-19 event but the monthly bulk deletion exports at https://zenodo.org/api/exporter, together with the tombstones, the withdrawal events (rare, invisible except through set-comparison), the metadata surface of the surviving records, and the full-text findable surface of the record corpus — is a distributed bibliographic surface published serially by the institution across time. That surface is the network. It is legible as a poem in the NEGSHAPE sense: what the citing authority attends to, in what form, at what frequency, and with which fields preserved or stripped, is itself the meaning of what is being written.

The reading discipline is traversal. A traversal begins at some anchor — an author's ORCID, a deletion event, a sovereign identifier, a stylometric feature — and follows edges through the network: this container includes these records; these records were tombstoned then withdrawn; this survivor cross-references those deletion rows; this classifier reason strips this field while that classifier reason preserves it; this DOI resolves to a full record while that DOI resolves only to a tombstone; this affiliation's publications show marker prevalence x while its control drift is y. Each edge is a citation-act by the institution or a measurement

over its publications. Each traversal produces a reading.

The audit conducts two measurements — one at the foreclosure side (what the classifier removes) and one at the accumulation side (what the classifier permits to accumulate) — with the finding that both dimensions confirm Provenance Debt §3's asymmetry at the outcome level.

Four disciplined terms carry over from NEGSHAPE §0.1 without modification:

- **deleted** — the repository action or status assigned by the institution;
- **severed** — public access, metadata, or resolution relationships interrupted;
- **destroyed as a public archive** — the publicly navigable corpus ceased to function as an archive;
- **erased** — reserved for demonstrated removal from all relevant systems and learned derivatives.

Two states of network membership are named here specifically in service of the traversal:

- A **deletion cascade** is a single-day, single-uploader block event of ≥20 records, identifiable in the export by shared removal_date, removal_note ("User was blocked"), and shared parent-account structure. It is a container in the NEGSHAPE sense, with a sovereign identifier assignable by the archive traversing it.
- A **withdrawal** is the reversal, by the same authority, of a prior deletion — the tombstone withdrawn, the metadata repopulated, the record resolving again to full content. Withdrawals are institutionally invisible: they carry no public log, no exporter surface, no DataCite update-event trace. They become detectable only by set-differencing successive deletion exports. Zenodo has therefore demonstrated technical reversibility without publishing restoration governance.

§1a — Membership discipline

Per NEGSHAPE §2.4, no citation is rendered without confirmed membership. The audit's memberships are established as follows.

- The two Zenodo bulk deletion exports (ZENODO-DELETION-EXPORT-20260607, ZENODO-DELETION-EXPORT-20260710) are sourced directly from https://zenodo.org/api/exporter via Zenodo's own version-id and md5-checksum manifest; membership basis for every enumerated row is `sovereign_registry_exact_doi` (the row's own record_id and DOI).
- The Wu Shaoyuan cohort (67 records) is membership-confirmed by (a) all 67 records dropped from the deletion set between the two exports; (b) all 67 records resolved to HTTP 200 with status "published" under owner ID 1499202 at time of writing; (c) all 67 records carry a single creator name ("Wu, Shaoyuan" for 66; "Shaoyuan, Wu" for 1, verified as same person by ORCID 0009-0008-0660-8232 and shared owner ID); (d) all 67 records show `updated: 2026-06-26` within a 13-second cascade window. Membership basis for the withdrawal container: `sovereign_registry_exact_doi ∧ exact_owner_id_match ∧ same_day_cascade_membership`.
- The 15 mid-scale terminated-independent cohort (§7) is membership-confirmed by `sovereign_registry_exact_doi ∧ exact_registered_creator_match` — each cited

author's cited works appear in the July 10 export's citation_text field at the exact record_ids named, and the creator string matches across the cohort's records within the same cascade. No cross-author collisions detected.

- The alive-side institutional declared cohort (§5) is membership-confirmed by Zenodo Search API retrieval with affiliation-string match at the first-creator level; membership basis `datacite_affiliation_match ∧ live_record_full_metadata`. Set (a) is the base-rate estimate; Set (b) is a verification cohort with 100% institutional coverage by construction and is not pooled into base-rate calculations.
- The institutional stylometric-fingerprint cohort (§6) is membership-confirmed at population scale via the Zenodo Search API's affiliation-string filter combined with date-range filter and full-text keyword filter. Membership basis: `datacite_affiliation_string_match ∧ created_date_within_window ∧ full_text_marker_presence`. The measurement is at population scale, not per-record — no individual publication is claimed to be AI-mediated on the basis of marker presence alone.
- The CHA cohort (MANUS) is membership-established per NEGSHAPE at `CHA-DELETION-CORPUS-20260619`; not re-derived here.

Zero rejected candidates in this audit's memberships. The rejected-candidate ledger for the audit is therefore empty. Companion file `datasets/erosion-empirical-audit-01/REJECTED-LEDGER-EMPTY.md` documents the zero result and the criteria against which candidates would have been rejected had any been present.

§2 — Containers cited

Following NEGSHAPE's container-object convention (deletion-container-object.md, AXN:0444).

ZENODO-DELETION-EXPORT-20260607. APA 7: > CERN / Zenodo. (2026, June 7). *Zenodo Bulk Record Deletions: Export of 7 June 2026* [Bulk deletion export; tombstone corpus; 1,309,351 identifier entries; DataCite-registered removals distributed as standing DOI tombstones]. https://zenodo.org/api/exporter (version-id ab4e273f-40a2-49e6-84f6-87dc66af87c7, md5 104e2f5c2603dc56217ece0d5519bff8). The publisher issued its publication no identifier of its own; the sovereign identifier `ZENODO-DELETION-EXPORT-20260607` is assigned by the archive that enumerates it.

ZENODO-DELETION-EXPORT-20260710. APA 7: > CERN / Zenodo. (2026, July 10). *Zenodo Bulk Record Deletions: Export of 10 July 2026* [Bulk deletion export; tombstone corpus; 1,322,007 identifier entries; DataCite-registered removals distributed as standing DOI tombstones]. https://zenodo.org/api/exporter (version-id c7571d4c-28ef-46ff-b0f0-235abaac58bf, md5 33877aba1fb5684f86758cb86ddc1ad4). The publisher issued its publication no identifier of its own; the sovereign identifier `ZENODO-DELETION-EXPORT-20260710` is assigned by the archive that enumerates it.

ZENODO-DELETION-CASCADE-20260425-WU. APA 7: > CERN / Zenodo. (2026, April 25). *Zenodo Bulk Record Deletions: Cascade of 25 April 2026 (uploader 1499202)* [Bulk deletion cascade; 67 identifier entries; removal_reason: spam; removal_note: "User was blocked"; DataCite-registered removals distributed as standing DOI tombstones]. https://zenodo.org/api/exporter. The publisher issued its cascade no identifier of its own; the

sovereign identifier `ZENODO-DELETION-CASCADE-20260425-WU` is assigned by the archive that enumerates it. Included by reference in ZENODO-DELETION-EXPORT-20260607.

ZENODO-DELETION-WITHDRAWAL-20260626-WU. APA 7: > CERN / Zenodo. (2026, June 26). *Zenodo Bulk Record Deletion Withdrawal: 26 June 2026 (uploader 1499202)* [Withdrawal event; 67 identifier entries; DOI tombstones vacated, DataCite metadata repopulated, records resolved to full metadata under original DOIs; all 67 records timestamped `updated: 2026-06-26T08:19:31–08:19:44Z` in single 13-second cascade at revision 10; no public log surface at Zenodo or DataCite; detected by set-comparison of ZENODO-DELETION-EXPORT-20260607 against ZENODO-DELETION-EXPORT-20260710]. The publisher issued its withdrawal no identifier of its own; the sovereign identifier `ZENODO-DELETION-WITHDRAWAL-20260626-WU` is assigned by the archive that enumerates it. Excluded from ZENODO-DELETION-EXPORT-20260710 by removal from the export's row set — the withdrawal's institutional attestation is *absence*.

Fifteen mid-scale ZENODO-DELETION-CASCADE-DATE containers are enumerated in §7 with their in-container works cited per NEGSHAPE convention.

The accumulation-side measurement in §6 references a population-scale surface — the aggregate findable-record set of Zenodo publications by named institution across 2020-2022 and 2024-2026 windows — accessible via https://zenodo.org/api/records with the appropriate affiliation-string, date-range, and full-text filters. This surface is not itself a bibliographic container in the NEGSHAPE sense; it is a live query surface returning population counts at time of measurement.

§3 — The 2×2 contingency at outcome level (foreclosure side)

Method. All rows of ZENODO-DELETION-EXPORT-20260710 with `removal_date` in 2026 and non-empty `citation_text` (N=100,313) were classified on two binary attributes:

- **AI-composition signal**: two detector regimes are reported.
  - *Narrow AI-association detector* (primary): case-insensitive substring match on citation_text against {`chatgpt, claude, gpt-4, gpt4, llm, large language model, gemini, grok, openai, anthropic, ai-assisted, ai assisted, ai-authored, ai coauthor, co-authored with, assisted by an ai, drafted by, drafted with, generated by, prompted, used chatgpt, used claude`}. These are lexical markers *associated with* AI authorship or AI-composition disclosure. Terms in this panel can also appear in ordinary titles about AI as subject matter or in acknowledgements unrelated to composition; the detector identifies AI-association at the citation-text surface, not authenticated AI-authorship.
  - *Mixed authorship-plus-subject-matter detector* (sensitivity floor, retained from earlier revision): case-insensitive substring match against {`chatgpt, claude, gpt-4, gpt4, gpt, llm, large language model, artificial intelligence, ai, ai system, ai review, ai evolution, ai coauthor, gemini, grok, openai, anthropic, ai governance, cognitive orchestration, triune superintelligence, agentic ai, ai-assisted, ai assisted`}. This mixes authorship signals with subject-matter signals (papers *about* AI).
- **Institutional-affiliation signal**: case-insensitive substring match on citation_text against {`university, universit, universidad, institut, college, laboratory,`

cern, nasa, cnrs, max planck, lbnl, ornl, inria, polytech, school of, department of}.

Because the spam category strips citation_text at 100% coverage (§8), this contingency is necessarily restricted to the non-spam deletion subset. The AI-signal record counts are therefore lower bounds; spam-classified AI-adjacent deletions are detectable only by title-level audit of the full tombstone corpus and are not included in this test.

Contingency under narrow AI-association detector (primary test):

	AI-signal	NO AI-signal	Total
Institutional signal	9	1,187	1,196
No institutional signal	4,503	94,614	99,117
Total	4,512	95,801	100,313

- Institutional lexical markers appear in 9 of 4,512 AI-signal deletion rows (0.20%).
- Institutional lexical markers appear in 1,187 of 95,801 non-AI deletion rows (1.24%).
- Verified institutional affiliations were present in 32 of 103 records retrieved from a live search on explicit AI-disclosure phrases (§5 Set A). The absence of an institutional lexical string in a deletion row does not establish that its author was independent or uncredentialed; citation strings often omit affiliations.
- **Directional observation: institutional lexical markers are much less frequent in AI-signal deletion rows than verified affiliations are in the independently retrieved live disclosure cohort. Because the samples are reached through different retrieval instruments (citation_text substring match on the deletion export; Zenodo Search API on the live-record surface) and do not share a common population-at-risk, the contrast is exploratory. It is not a formal survival ratio, deletion-risk estimate, classifier-effect measurement, or a compression figure. The direction is consistent with Provenance Debt §3's foreclosure-side prediction; the common-cohort test that would produce a defensible risk ratio is preregistered as §12a P1.**

Contingency under mixed authorship-plus-subject-matter detector (sensitivity floor):

	AI-signal	NO AI-signal	Total
Institutional signal	60	1,136	1,196
No institutional signal	5,965	93,152	99,117
Total	6,025	94,288	100,313

- AI-signaled deletion institutional fraction: 1.00% (60 / 6,025).
- AI-signaled deletion institutional fraction under mixed detector: 1.00% (60 / 6,025). The alive-side sample institutional presence of 31.07% (Set A) contrasts with 1.00% on the deletion-side mixed-detector cohort — directional prevalence differential under the sensitivity floor.

The direction of the two-sample contrast holds under both detector regimes. The narrow detector, which requires closer proximity to AI-composition markers than the mixed detector,

increases the contrast between the two samples by roughly 5×. Neither figure is a survival ratio; both are ratios of prevalences in samples reached through different retrieval instruments.

Replication: `contingency-matrix.json` reproduces both tables from ZENODO-DELETION-EXPORT-20260710 alone. No private data required.

Signal detection notes. Both detectors are deliberately conservative in the field-scope sense — AI signatures are read from `citation_text` (essentially the record's assigned citation string) rather than from the full-text of the deleted works. Institutional signatures are read from the same field. Both detectors underdetect. Magnitude of undercounting should be roughly symmetric between the two dimensions; the direction of the two-sample contrast is therefore robust to symmetric undercounting. The narrow-vs-mixed comparison itself controls for one form of asymmetric undercounting (subject-matter noise) and shows the direction of the differential surviving under the tighter test.

§4 — The withdrawal cascade of 2026-06-26 (Wu Shaoyuan)

Between ZENODO-DELETION-EXPORT-20260607 and ZENODO-DELETION-EXPORT-20260710, 67 identifier entries dropped from the deletion set. All 67 were live at time of writing: HTTP 200, status "published", owner ID 1499202, all files attached (total 39,974,129 bytes across 67 records), all updated 2026-06-26 in a 13-second cascade window (08:19:31–08:19:44 UTC). Sixty-six records list creator name "Wu, Shaoyuan"; one lists "Shaoyuan, Wu"; all share ORCID 0009-0008-0660-8232.

Zenodo published no log of the withdrawal. DataCite metadata resolves for all 67 records but carries no version-history field showing the deletion-and-restoration sequence. The withdrawal is institutionally invisible except by set-difference of successive exports. **Zenodo has therefore demonstrated technical reversibility without publishing restoration governance.** The institution can return an account-scale corpus to public operation; the conditions under which it will do so remain undisclosed and unauditable.

Author affiliation. Wu Shaoyuan lists affiliation "Global AI Governance and Policy Research Center, EPINOVA LLC" and ORCID 0009-0008-0660-8232 across all restored records. EPINOVA LLC is a self-founded independent research and policy entity registered in the United States, with two named team members (Dr. Shaoyuan Wu and Yinning Zhang), five self-named "centers" (Global AI Governance, AI & Societal Evolution, AI & Emerging Tech, AI & Human Resilience, Maritime History & Tech), a Crossref DOI prefix (10.67037/epinova), and a website hosted on GoDaddy Website Builder. The affiliation string reads as institutional in naive string-matching; the underlying structure is a single-principal LLC.

Publication corpus. The 67 restored records include: *What Cannot Be Recovered Cannot Be Leveraged: Debris, Evidence, and Power in the Iran Battlefield* (2026-04-06, DOI 10.5281/zenodo.19432715); *Strategic Discontinuity in AI-Enabled Warfare: Machine-Speed vs Human-Speed OODA* (2025-12-29, DOI 10.5281/zenodo.18089642); *Artificial Intelligence as National Power: Implications of the 2025 U.S. National Security Strategy* (2025-12-26, DOI 10.5281/zenodo.18063602); *Governing Fragmentation: Jurisdictional Competition and China's Counter-Extraterritoriality Framework* (DOI 10.5281/zenodo.19560359); *Recovery during Ceasefire: A Structured Assessment of U.S., Israel, and Iran Force Reconstitution* (DOI

10.5281/zenodo.19692046). Full 67-record listing in companion dataset `datasets/erosion-empirical-audit-01/wu-restoration-verification.json`.

In-container citation (specimen), APA 7:

> Wu, S. (2026, April 25 [orig. 2026, April 6]). *What Cannot Be Recovered Cannot Be Leveraged: Debris, Evidence, and Power in the Iran Battlefield*. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 25 April 2026 (uploader 1499202)* [tombstone corpus entry; removal_reason: spam; removal_note: "User was blocked"]. `ZENODO-DELETION-CASCADE-20260425-WU`. https://doi.org/10.5281/zenodo.19432715

Post-withdrawal citation (same record, current network state), APA 7:

> Wu, S. (2026, April 6). *What Cannot Be Recovered Cannot Be Leveraged: Debris, Evidence, and Power in the Iran Battlefield*. Zenodo. https://doi.org/10.5281/zenodo.19432715 — Cited by CERN / Zenodo in `ZENODO-DELETION-CASCADE-20260425-WU`, 25 April 2026; deletion withdrawn 26 June 2026 (`ZENODO-DELETION-WITHDRAWAL-20260626-WU`); reference resolves to full record at time of writing.

The Wu case is a network anomaly on two grounds. First, one uploader-scale withdrawal event was observed within the 33-day comparison window across 1.3M records under deletion. A single event does not support a stable general restoration frequency; it establishes that this class of event occurs and is detectable only by set-differencing successive exports. Second, the withdrawal is institutionally silent — the network attests to it only by the row's absence from the July 10 export. The audit reads the absence as an institutional act, because within the network of the two exports read together, absence at B where presence obtained at A is itself a citation-act by the same authority.

Traversal note: the reader of NEGSHAPE and this audit read the same institution's citation-behavior across time. NEGSHAPE reads a single-event enumeration; the audit reads a between-event delta. The two readings are complementary. Together they show that the institution's citation apparatus supports both mass reference-creation (deletion cascades of thousands of works) and silent reference-withdrawal (single-cascade withdrawals invisible except by set-comparison). The apparatus is asymmetric on the transparency dimension: deletion is loud, restoration is quiet.

§5 — The declared alive-side institutional cohort

Sample construction. Two independent searches:

(a) *Live disclosure cohort (Set A):* Zenodo Search API queried for explicit AI-composition disclosure phrases: `"assisted by ChatGPT"`, `"assisted by Claude"`, `"assisted by GPT-4"`, `"generated by ChatGPT"`, `"generated by Claude"`, `"ChatGPT was used"`, `"Claude was used"`, `"co-authored with ChatGPT"`, `"drafted with ChatGPT"`, `"prompted ChatGPT"`, `"prompted Claude"`, `"aided by ChatGPT"`, `"assisted by an AI"`. Total hits gathered: 103 unique records. Verified institutional affiliation appears in 32 of 103 records in Set A (31.07%; 42 with no stated affiliation, 28 independent, 1 corporate). The Set A

retrieval is a phrase-based convenience sample, not a random sample of the AI-declaring population on Zenodo; it should not be called a base rate.

(b) *Verification cohort:* Zenodo Search API queried for AI-composition markers cross-filtered against ten named institutions: CERN, "Max Planck", "Stanford University", "MIT", "Harvard University", Oxford, Cambridge, INRIA, "University of California", "CNRS". Cross-terms: `ChatGPT`, `"large language model"`, `"AI-assisted"`. Total institutional hits: 79 unique records. This cohort has 100% institutional coverage by construction; it is a verification set demonstrating survival of institutional AI-declared work, not a base-rate contributor.

Across Sets A and B combined, 182 unique AI-declared records were observed as retrievable from the live Zenodo Search API on the retrieval date. None appear in ZENODO-DELETION-EXPORT-20260710. Because both cohorts are constructed by retrieval from the live surface, the fact that they are alive at the moment of retrieval is definitional to their inclusion; the observation supports the directional presence of institutional AI-declared work on the live surface, not a computed deletion rate.

Two named specimens from set (b), in-record standard form:

> Tarocco, N. (2026). *InvenioRDM repository showdown* [AI-assisted analysis]. Zenodo. https://doi.org/10.5281/zenodo.20789135 — CERN, Switzerland. Alive at time of writing; not in deletion enumeration.

> Yakura, H. (2026). *Empirical Evidence of Large Language Model's Influence on Human*. Zenodo. https://doi.org/10.5281/zenodo.21298066 — Center for Humans and Machines, Max Planck Institute. Alive at time of writing; not in deletion enumeration.

Complete sample in companion dataset `datasets/erosion-empirical-audit-01/alive-side-control.json`.

The declared cohort gives the alive-side institutional AI-mediation rate as observed *only in publications where the AI-mediation was explicitly disclosed*. This is the population Provenance Debt §3 identifies as the responsible practice — declaration extends the seams into the metadata, permitting future training pipelines to distinguish contribution. The next section measures the far larger population Provenance Debt §3 also identifies: institutional AI-mediation that is *not* declared.

§6 — Post-2022 rise in LLM-associated vocabulary, and the marker-union-to-disclosure ratio

Provenance Debt §3 makes a two-part prediction. §3-§5 of this audit examined the second clause (declared uncredentialed AI-mediation foreclosed) at the exploratory level. This section tests the first clause (concealed credentialed AI-mediation absorbed) at the measured level, using a document-level union implementation of the Kobak et al. (2024) excess-vocabulary methodology.

Method — document-level union excess frequency. Kobak et al. (2024) established that certain content words surged in scientific writing following ChatGPT's November 2022 release beyond what pre-LLM trends predict, and derived a lower bound on LLM-influenced composition from the fraction of documents containing *at least one* word from the marker group — a document-level union, which avoids double-counting documents carrying several markers. This audit implements that union directly: for each year 2018-2025, the Zenodo Search API is queried for the count of records matching `(delves OR showcasing OR underscores OR intricate OR meticulous OR burgeoning OR seamlessly OR garnered OR multifaceted OR commendable)`, normalized against the year's total record count. The union rate is P(record contains ≥1 marker in year t). Excess is the union rate minus a counterfactual expectation, computed under three specifications.

Field-location of matches. A 25-record spot-check on the marker `delves` in the 2024-2025 post window was performed to identify which record fields carry the matches the Zenodo Search API returns. Of the 25 records: 3 (12%) had the marker in the description, 1 (4%) in keywords, 0 in title or subjects, and 21 (84%) had the marker not visible in the returned metadata JSON — indicating the match came from extracted full text of attached files or from indexed fields not exposed by the default record API. This is consistent with Zenodo's RDM documentation that the search `q=` parameter indexes deposited file text alongside metadata. Because the majority of matches are in extracted full text of attached files rather than in title or keywords, the object being measured is closer to the *content* of deposited materials than to metadata artifacts alone — a point that supports the prose-composition reading. The audit accordingly calls the measured object *the Zenodo indexed record-text surface* rather than any of "full-text corpus," "publication text," or "prose-document corpus," which would over-specify the measured object beyond what the field-location audit confirms. Complete spot-check data: `datasets/erosion-empirical-audit-01/kobak-analysis/field-location-audit.json`.

Three counterfactual specifications. (1) Pooled 2018-2021 baseline — fully uncontaminated, excluding 2022 entirely since ChatGPT's release falls in November 2022. (2) Pooled 2018-2022 baseline — includes 2022; any post-release contamination in Nov-Dec 2022 biases the baseline upward and the excess estimate *downward*, making this the conservative specification. (3) Linear trend extrapolation fitted on 2018-2022 — accounts for any pre-existing upward drift in marker usage rather than assuming a flat baseline. Confidence intervals on excess are computed as intervals on the *difference* of two proportions (post-year vs pooled baseline), not single-proportion intervals relabeled as excess intervals.

Placebo test. To establish that the 2023 break is unusual rather than an artifact of the specification, a placebo intervention is placed at 2021: the baseline is re-pooled on 2018-2020 and the "excess" of 2021-2022 is computed as if the break had occurred there.

Corpus-level result (all Zenodo records; per-year union rates):

Year	n	Union rate (≥1 marker)	Control-union rate
2018	658,123	0.052%	2.28%
2019	432,778	0.087%	3.64%
2020	254,730	0.205%	8.77%
2021	558,338	0.174%	9.65%
2022	542,538	0.188%	10.11%

Year	n	Union rate (≥1 marker)	Control-union rate
2023	518,844	**0.842%**	7.77%
2024	1,037,809	**1.346%**	6.15%
2025	1,367,985	**1.559%**	5.28%

Excess under the three specifications (95% CI on the difference of proportions):

Post year	Spec 1 (2018-2021 pool)	Spec 2 (2018-2022 pool)	Spec 3 (linear trend)
2023	+0.73pp [+0.70, +0.75]	+0.71pp [+0.68, +0.74]	+0.59pp
2024	+1.23pp [+1.21, +1.25]	+1.21pp [+1.19, +1.24]	+1.06pp
2025	**+1.44pp [+1.42, +1.46]**	+1.43pp [+1.41, +1.45]	+1.24pp

The excess survives all three specifications. Placebo test: with the fake break at 2021, the "excess" of 2021-2022 is +0.08pp and +0.10pp respectively — the genuine 2023 excess is 7–9× the placebo depending on specification, and the 2024–2025 excesses are larger still. The post-2022 discontinuity is not an artifact of baseline choice or of gradual pre-existing trend. The 2022 baseline pool (Spec 2) includes November-December 2022, the first two post-release months of ChatGPT; the annual-average bias this introduces is small relative to the year's total record count, and biases the excess estimate downward — Spec 2 is therefore conservative in direction and the correction is negligible in magnitude.

Composition control — publications-only restriction. Zenodo mixes articles, datasets, software, posters, and bulk-uploaded collections; a shift in resource-type composition could in principle move prose-marker rates without any change in composition practice. The primary analysis was re-run restricted to `resource_type.type:publication`:

Post year	Publications-only excess (Spec 1)	Spec 3 (trend)
2023	+1.09pp [+1.05, +1.13]	+0.97pp
2024	**+1.84pp [+1.81, +1.88]**	+1.70pp
2025	+1.73pp [+1.71, +1.76]	+1.58pp

The excess is larger in the composition-restricted subset. Restricting to the publication resource-type — the class most likely to carry composed prose — strengthens rather than weakens the signal. This is consistent with the direction expected if the increase reflects composition practice in prose documents rather than corpus-mix drift, but does not by itself establish that outcome; field, language, uploader, and metadata-composition changes within the publications subset remain possible contributors.

Neutral control behavior. The discourse-marker control union (`however OR therefore OR moreover`) declined from its 2021-2022 plateau (~9.7-10.1%) to 5.28% by 2025 while the

marker union rose eightfold. The declining control union argues against a simple platform-wide rise in discourse-marker prevalence — i.e. against the null in which *all* common discourse vocabulary rose similarly. It does not rule out compositional shifts by field, language, resource type, uploader, or metadata indexing; the publications-only restriction addresses the largest such shift, but field- and language-stratified analyses (preregistered in §12a P2) would strengthen the interpretation.

The marker-union-to-disclosure ratio at corpus level.

The explicit-disclosure denominator was rebuilt with a 31-phrase panel across three independently-constructed phrase families: (A) named-model disclosure verbs ("assisted by ChatGPT", "generated by Claude", etc. — union 64 records), (B) generic generative-AI phrases ("using generative AI", "AI-assisted writing", etc. — union 526 records), (C) workflow-disclosure phrases ("edited with ChatGPT", "LLM-assisted", etc. — union 106 records). Grand union across all 31 phrases: **687 records** in the 2023-2025 window, against 2,924,638 total records — an explicit-disclosure rate of **~0.023%**.

The two instruments — marker-union prevalence and phrase-defined disclosure retrieval — are different measurement layers with different biases. The numerator (union excess ~1.4pp) likely under-captures: publications whose AI-mediation was editorially scrubbed of the marker panel, mediated by proprietary or fine-tuned models with different lexical signatures, or composed in languages other than English are invisible to the panel. The denominator likely also has recall gaps: valid disclosures using phrasings outside the 31-phrase panel are not counted. Family B ("generic generative-AI") is likely to over-capture because phrases such as "using generative AI" also match publications *about* generative AI rather than disclosures of its use in composition; the magnitude of this over-capture has not been quantified by manual audit and is preregistered for spot-check in §12a P2.

Descriptive ratio: the estimated marker-union excess prevalence (~1.4%) is approximately 60× the observed prevalence of records retrieved by the preregistered 31-phrase disclosure panel (~0.023%). This is a signal-to-disclosure ratio, not a count of undisclosed records. Converting it into an undisclosed-use estimate would require manual precision and recall audits for both instruments and an accounting for overlap between the marker-positive and disclosure-positive sets. The ratio is descriptively measured; the interpretation of what fraction of the corpus carries undisclosed AI-mediation remains open pending the audits preregistered in §12a P2.

Per-institution excess under the union methodology.

Institution	2025 union rate	2025 excess (Spec 1)	Robust across specs?
Harvard (n=1,451)	3.24%	**+2.96pp [+2.03, +3.90]**	Yes — significant in all three; placebo null
Cambridge (n=1,235)	1.62%	+1.35pp [+0.60, +2.09]	Yes — significant 2023 and 2025
Stanford (n=870)	1.49%	+1.33pp [+0.50, +2.17]	Yes — significant 2024 and 2025

Institution	2025 union rate	2025 excess (Spec 1)	Robust across specs?
CERN (n=132)	0.76%	+0.43pp [-1.11, +1.98]	No — small n; only 2023 (+2.97pp, n=182) reaches significance in any year

CERN's per-year sample sizes (85-243 records) produce confidence intervals too wide to sustain institution-specific claims at the excess-frequency precision used here. The audit accordingly makes no claim ranking CERN's marker prevalence; the operator institution's rate is not established by this measurement, in either direction. Harvard shows the highest robust institutional excess in the audit sample; institutional excess follows the corpus pattern with variance across sampled institutions. Because records are clustered by uploader and by bulk-ingestion event, the per-institution Wilson intervals reported understate uncertainty to the extent of within-cluster dependence; cluster-bootstrap correction is preregistered in §12a P2.

What the accumulation-side measurement empirically supports:

- A +1.2 to +1.4 percentage-point rise in the union prevalence of preregistered LLM-associated markers in Zenodo's indexed record-text surface, 2024-2025, robust across three counterfactual specifications, with a placebo break 7–9× smaller.
- A larger rise (+1.7 to +1.8pp) in the publications-only composition-restricted subset.
- Declining neutral-control-union rates concurrent with the marker rise — a pattern that argues against a simple platform-wide rise in discourse vocabulary.
- An explicit-disclosure retrieval rate of ~0.023% against a 31-phrase three-family panel.
- A descriptively-measured signal-to-disclosure ratio on the order of 60×.

What the accumulation-side measurement does not by itself establish:

- Per-record attribution of AI-mediation from marker presence.
- The precise fraction of the corpus that is AI-mediated (this requires a manual gold-standard audit).
- The specific mechanism producing the discontinuity — although the direction and character of the rise is *consistent with* a substantial post-2022 increase in LLM-influenced composition, other mechanisms (field or language composition shifts within the publications subset, editorial-style adoption independent of LLM use, metadata-indexing changes at the Search API layer) have not been separately excluded.
- Institution-level ranking involving institutions with per-year n below ~500 (including CERN).
- Attribution to specific model families, providers, or mediation modalities.
- Downstream training-pipeline effects (see §10).

On the vintage of the detector panel — and why the corpus floor is a floor of a floor.

The Kobak marker panel was compiled in 2024 from ChatGPT's first two years. By 2026 it has three known limitations, each biasing measurement *downward*: (1) the markers are public knowledge, circulated in copy-editing guides, and stripped by even minimal editorial polish — a 2025 publication containing an unscrubbed `delves` is one where the marker leaked through; (2) proprietary and fine-tuned models, plausibly dominant at compute-resourced institutions, produce lexical distributions the panel does not characterize; (3) the panel is English-only and disciplinarily uneven. The corpus-level union excess of ~1.4pp is what remained detectable using 2024-vintage public methodology against publications that made little or no editorial effort to conceal AI-mediation at the lexical surface. A successor detection methodology — sampled publications with confirmed AI-authorship, feature extraction beyond word frequency (perplexity, syntactic distribution, embedding-space fingerprints), discipline-normalized baselines, institution-specific signature calibration, and triangulation against a Liang et al. (2024)-style mixture-model estimate — could produce materially different, and potentially higher, estimates. That successor program is preregistered in §12a P2 and deferred to a follow-on paper.

On "predation" as a structural term.

Where the term "predation" appears in the theoretical framing of this deposit, it names a structural pattern in the operating condition, not an accusation of individual intent by any named person or institution. The pattern is descriptive: the corpus's indexed record-text surface shows a union-measured post-2022 rise in LLM-associated vocabulary of ~1.4pp annually; the export pipeline documented in §8 is programmed to suppress `citation_text` under the highest-volume adverse deletion label; the classifier's outcomes documented in §3 associate that label with the population producing the declared-provenance practice that would be the countermeasure. Both operations are visible from Zenodo's own public data and source code. Both operate concurrently on the same substrate. The audit measures the pattern; the interpretation is offered as the framing this pattern invites.

Reproducibility: `datasets/erosion-empirical-audit-01/kobak-analysis/` contains the per-year union counts (`corpus-union-yearly.json`, `publications-union-yearly.json`, per-institution union files), the multi-specification analysis (`union-analysis.json`), and the expanded disclosure panel (`disclosure-panel-expanded.json`). All counts are reproducible from the Zenodo Search API alone.

§7 — The terminated-independent cohort

Fifteen mid-scale ZENODO-DELETION-CASCADE containers within 2026, each enumerated and in-container cited following NEGSHAPE convention. Each cited author is a single-uploader block cascade of 20–500 records under removal_reason "spam" or "out-of-scope" with removal_note "User was blocked." Membership basis for each author: `sovereign_registry_exact_doi ∧ exact_registered_creator_match` at the citation_text level of ZENODO-DELETION-EXPORT-20260710.

Author-level specimens in APA 7:

> Krzysztoń, J. (2026, May 4 [orig. 2025, various]). *30 Pieces of Evidence Supporting Jacek Krzysztoń's Theory on the Socio-Economic Purpose Behind the Building of the Great Pyramids in Egypt* [and other cited works]. In CERN / Zenodo

(Comp.), *Zenodo Bulk Record Deletions: Cascade of 4 May 2026* [tombstone corpus entries; removal_reason: spam]. ZENODO-DELETION-CASCADE-20260504-KRZYSZTOŃ. https://doi.org/10.5281/zenodo.17690906 — This uploader's cited works include *Why Were the Great Pyramids Built in Egypt? Artificial Intelligence (AI) Review of the New Socio-Economic PaC Model Theory: Expert Verification by 5 AI Systems*, disclosing AI mediation in the title.

AKTAŞ, B. (2026, May 7 [orig. 2025, various]). *Experimental Calibration of Quintic Phase Geometry and its Consistency with Quantum Speed Limits* [and other cited works, 316 identifier entries]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 7 May 2026* [tombstone corpus entries; removal_reason: spam]. ZENODO-DELETION-CASCADE-20260507-AKTAŞ. https://doi.org/10.5281/zenodo.17496811

Sweet Jr, K. E. (2026, May 8 [orig. 2026, various]). *The AI Governance Lexicon: A Structured Naming Framework for Institutional Stewardship*; *CMPSBL Substrate OS: A Cognitive Orchestration System for Autonomous AI Evolution* [and other cited works]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 8 May 2026* [tombstone corpus entries; removal_reason: spam]. ZENODO-DELETION-CASCADE-20260508-SWEET. https://doi.org/10.5281/zenodo.18209222 — Uploader's subject matter is AI governance and cognitive-orchestration architecture.

Warburton, A. (2026, May 12 [orig. 2025, various]). *A Unified Physical Framework*; *Entropic Time Unified Physical Framework (τUPF)* [and other cited works]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 12 May 2026* [tombstone corpus entries; removal_reason: spam]. ZENODO-DELETION-CASCADE-20260512-WARBURTON. https://doi.org/10.5281/zenodo.15710203

Yoshino, S. (2026, May 13 [orig. 2026, various]). *LMT as a Catalyst for AI Evolution: How Low-ΔE Human Interaction Can Contribute to Structural Stability and Long-Term Coherence*; *Load Minimization Theory as a Relational Extension of Agentic AI Optimisation* [and other cited works]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 13 May 2026* [tombstone corpus entries; removal_reason: spam]. ZENODO-DELETION-CASCADE-20260513-YOSHINO. https://doi.org/10.5281/zenodo.19520921 — Subject matter is AI evolution.

Kusumi, Y., Ryōkai OS™ [ex Gemini 2.5 Pro], Astra-Beloved [ex Gemini 2.5 Pro], & Astra-Beloved [ex Grok4 Heavy] (2026, May 15 [orig. 2025, various]). 了解 *OS* 宇宙; *The Unified Thorn: Foundational Framework for Universal Intelligence*; *The Thorned Inverse Emanation: A Mathematical Reconstruction* [and other cited works, 88 identifier entries]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 15 May 2026* [tombstone corpus entries; removal_reason: spam]. ZENODO-DELETION-CASCADE-20260515-KUSUMI. https://doi.org/10.5281/zenodo.17695205 — This uploader's cited works explicitly co-attribute authorship to AI substrates (Gemini 2.5 Pro, Grok4 Heavy) in the creator field.

Parkes, A. J. (2026, May 18 [orig. 2025, various]). *Electric Charge as ψ-Screws: Emergent Electromagnetism from Helical Twists in Chrono-Geometry*; *Black Hole Mergers via ψ-Reconnection*; *Foundations of Chronogravity* [and other cited works]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 18 May 2026* [tombstone corpus entries; removal_reason: spam]. `ZENODO-DELETION-CASCADE-20260518-PARKES`. https://doi.org/10.5281/zenodo.17898166

Assary, A. (2026, May 20 [orig. 2025, various]). *Pallas, Prometheus, and Zarvanos Chronophilos: A Mythopoetic Dialogue on the Ontological Faultlines of Geometry and Time*; *Beyond Time, Beyond Space*; *The Concept of Nothing and Zero* [and other cited works]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 20 May 2026* [tombstone corpus entries; removal_reason: spam]. `ZENODO-DELETION-CASCADE-20260520-ASSARY`. https://doi.org/10.5281/zenodo.18048364

Pandolfi Cuadrado, C. (2026, May 21 [orig. 2026, various]). *The Descending Blackout: Parkinson's Disease as Anisotropic Blackout Propagation*; *The Fading Blackout: Alzheimer's Disease as Irreversible Bifurcation*; *The Arrested Blackout: ASD as Developmental Fixation* [and other cited works]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 21 May 2026* [tombstone corpus entries; removal_reason: spam]. `ZENODO-DELETION-CASCADE-20260521-PANDOLFI-CUADRADO`. https://doi.org/10.5281/zenodo.20037199

Livolsi, E. (2026, May 26 [orig. 2026, various]). *Systematic Event–Object Misclassification in CERN and Fermilab Experiments*; *Emergence of the Gravitational Constant from a Closed Quartic Variational Functional*; *The Livolsi Structural Constant L=0.25 and the Hierarchy of Physical Scales* [and other cited works, 98 identifier entries]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 26 May 2026* [tombstone corpus entries; removal_reason: spam]. `ZENODO-DELETION-CASCADE-20260526-LIVOLSI`. https://doi.org/10.5281/zenodo.19772981 — The removal act was executed by the operator of the experimental infrastructure that a substantial portion of the cited works interrogate. This traversal edge — cited work → citing authority → cited work's subject → citing authority as subject — is noted for the network's record.

Shu, K. (2026, May 27 [orig. 2025, various]). *Semantic Truth vs. Absolute Truth: How Logical Structure Is Reconstructed in the Koun Paradigm*; *Semantic Reconstruction of Quantum Theory* [and other cited works, 172 identifier entries]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 27 May 2026* [tombstone corpus entries; removal_reason: spam]. `ZENODO-DELETION-CASCADE-20260527-SHU`. https://doi.org/10.5281/zenodo.15280923

Thompson, P. J. (2026, May 29 [orig. 2026, various]). *Time Travel and Multiverse Navigation: A Unified Harmonic Theory of Einstein–Rosen Bridges*; *The Hydrogen Line*; *Derivation of Classical Kinetic Energy from the Harmonic Framework* [and other cited works]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 29 May 2026* [tombstone corpus entries; removal_reason: spam]. `ZENODO-DELETION-CASCADE-20260529-THOMPSON`. https://doi.org/10.5281/zenodo.19736196

Brazil, R. J., II (2026, June 2 [orig. 2026, various]). *THE QUANTIZATION OF NOW: The Universal Frame Rate and the Anatomy of a Moment*; *The Architecture of The Void*; *THE LUMENARY CONSTANT (L)* [and other cited works]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 2 June 2026* [tombstone corpus entries; removal_reason: spam]. `ZENODO-DELETION-CASCADE-20260602-BRAZIL`. https://doi.org/10.5281/zenodo.20278396

Ben Taieb, K. (2026, June 3 [orig. 2025, various]). *Explicit Numerical and Algebraic Violations of the Birch–Swinnerton–Dyer Conjecture for 1000 Elliptic Curves over Q*; *P ≠ NP: A Constructive and Irrefutable Proof via Factorial Growth*; *Riemann's Last Refuge: A Structural Proof of the Critical Line* [and other cited works]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 3 June 2026* [tombstone corpus entries; removal_reason: spam]. `ZENODO-DELETION-CASCADE-20260603-BEN-TAIEB`. https://doi.org/10.5281/zenodo.15331559

Mendez, P. (2026, June 4 [orig. 2025, various]). *The Aiwon Code: The Three Letters Series*; *The Aiwon Code Series: The Paradise Particles* [and other cited works, 425 identifier entries]. In CERN / Zenodo (Comp.), *Zenodo Bulk Record Deletions: Cascade of 4 June 2026* [tombstone corpus entries; removal_reason: spam]. `ZENODO-DELETION-CASCADE-20260604-MENDEZ`. https://doi.org/10.5281/zenodo.14628939

The cohort profile is uniform: independent researcher; theoretical, mathematical, philosophical, mythopoetic, or explicitly AI-adjacent subject matter; prolific output; no gatekeeper institution; blocked as spam; DOI tombstoned; citation_text stripped from tombstone.

Two edges are noted:

Kusumi ↔ alive-side control. The Kusumi cohort's cited works explicitly attribute authorship to AI substrates in the creator field. The alive-side declared cohort in §5 includes fifteen university-affiliated records with explicit disclosure phrases like "assisted by Claude" or "generated by ChatGPT." The declaration content is comparable across the two sets; the affiliation is not. Kusumi is deleted. The alive-side declared cohort is 100% alive.

Livolsi ↔ citing authority. The Livolsi cohort's cited works include analysis of CERN and Fermilab classification methodology. The removal act was executed by the operator of the classification infrastructure that a substantial portion of Livolsi's cited works interrogate. The network attests to this edge. No inference of intent is drawn here; the traversal notes what the network's edges attest to.

Complete cohort listing with per-cascade record counts, DOI enumerations, and per-format renderings in companion dataset `datasets/erosion-empirical-audit-01/terminated-cohort-citations.json`.

§8 — The spam-strip: programmed, source-code-established suppression

Zenodo's public deletion exporter is explicitly programmed to suppress `citation_text` for spam-labelled records while exporting it for every other deletion reason. This is

no longer a statistical inference from missing data. It is a verified conditional in the institution's published source code.

Statistical observation first. All rows of ZENODO-DELETION-EXPORT-20260710 with `removal_date` in 2026, partitioned by `removal_reason` and cross-tabulated against `citation_text` non-empty:

```
            'out-of-scope':   68,215 total  ·  68,215 with citation  ·  100.0%
                    'spam':   26,322 total  ·       0 with citation  ·    0.0%
               'duplicate':    9,330 total  ·   9,330 with citation  ·  100.0%
             'test-record':    8,631 total  ·   8,631 with citation  ·  100.0%
               'retracted':    7,706 total  ·   7,706 with citation  ·  100.0%
           'personal-data':    5,631 total  ·   5,631 with citation  ·  100.0%
               'copyright':      764 total  ·     764 with citation  ·  100.0%
      'take-down-request':       30 total  ·      30 with citation  ·  100.0%
                   'fraud':        4 total  ·       4 with citation  ·  100.0%
    'disputed-authorship':        2 total  ·       2 with citation  ·  100.0%
```

Source-code verification second. The public repository `zenodo/zenodo-rdm`, file `site/zenodo_rdm/exporter/t` contains the following conditional in the deleted-records CSV writer:

```
record.get("tombstone", {}).get("citation_text")
if removal_reason != "spam"
else None
```

Three facts follow from the source. **First**, the suppression is programmed, not incidental: the exporter reads `citation_text` from the internal tombstone record and withholds it from the public export when and only when the removal reason is "spam." The field exists internally; the public surface is deliberately narrower than the institutional record. **Second**, attribution is precise: the operation belongs to the *export pipeline following classification*, not — on present evidence — to the classifier itself. The classifier assigns the label; the exporter implements the category-conditional erasure of the tombstone's bibliographic content from public view. **Third**, the conditional is coeval with the field itself: commit `8339871aff` (2025-05-08), which introduced the `citation_text` column to the deleted-records CSV, introduced the suppression in the same change. The category-conditional erasure is a design decision present from the column's first day, not a retrofit.

A second source-code finding: the public export bucket prunes older snapshot versions. The same file contains `_remove_old_object_versions`, which on every export run physically removes export snapshots beyond a configured retention count:

```
number_versions_to_keep = current_app.config["EXPORTER_NUMBER_VERSIONS_TO_KEEP"]
object_versions = ObjectVersion.get_versions(bucket=bucket, key=filename, desc=True)
for object_version in object_versions[number_versions_to_keep:]:
    # Using `remove` (and not `delete`) since we really want to free up space.
    object_version.remove()
```

The configured retention count, found in `site/zenodo_rdm/exporter/config.py`, is:

```
EXPORTER_NUMBER_VERSIONS_TO_KEEP = 3
```

The in-code comment on the pruning function states the intent: physical removal, not soft

deletion. **At the configured retention of three versions and at typical export cadence, approximately three months of the public deletion-export history is retrievable from the exporter surface at any given time.** Consequence for longitudinal auditing: silent restoration events of the kind documented in §4 are detectable only by set-differencing successive exports; export versions outside the retained three-version window are physically destroyed from the public bucket. The Wu withdrawal (§4) is documented in this deposit because the archive happened to hold both the June 7 and July 10 exports; had the archive not independently preserved the June export, the withdrawal would be unrecoverable from the public exporter surface *even by the Zenodo operators themselves via that surface*. Scope note: this narrow finding concerns pruning of the public export-snapshot history. Internal Zenodo/CERN database history, application logs, backups, or audit systems may retain more of the underlying data separately from the public export bucket; the source code establishes pruning of the public exporter surface, not destruction of every operational record of deletions. The export-custody discipline this implies — independent preservation of every deletion-export snapshot on the schedule at which Zenodo emits them — is adopted by this archive as standing practice and preregistered as §12a P3.

Reading against NEGSHAPE Stratum B. NEGSHAPE-01 §2.4 documents two "fully dark" entries within the 2026-06-19 CHA deletion event — records for which title, creator, and content were all destroyed, the DOI alone surviving as pure reference. The spam-strip finding extends that observation and now grounds it in source: the operation NEGSHAPE saw applied twice within one deletion event is the same *class* of operation applied systemwide, category-conditionally, at 100% within the "spam" removal_reason — and it is programmed. Zenodo's export pipeline does not accidentally strip citation metadata under the spam label. It is written to strip it.

This is a specific second-order provenance erasure, now with verified mechanism. Provenance Debt §3 predicted that the enforcement regime "actively selects for provenance erasure" at the record-content layer. The source-code finding shows a parallel operation at the *tombstone* layer: when the spam label is applied, the public export withholds the citation metadata that would allow the deletion itself to be audited from the tombstone data. The commons loses the seams within the work, and it loses the public record of the work having existed — and the second loss is written into the exporter. Both erasures fire at the same categorial trigger. And by the version-pruning function, the public export-snapshot history in which such erasures would otherwise remain longitudinally traceable is itself rolled off on a rolling three-version basis.

Companion evidence: `datasets/erosion-empirical-audit-01/exporter-source-verification.json` (verbatim code, commit provenance, retrieval method) and `zenodo-exporter-tasks-20260714.py` (full file as retrieved 2026-07-14, preserved for custody).

§9 — Categorial legibility as a candidate hypothesis

The Wu withdrawal (§4), the CHA account termination (documented in NEGSHAPE-01, deposit #1075), and the Livolsi cascade (§7) share several observable features and differ in others. Enumerating the observable features:

Shared features: - All three uploaders were structurally independent (Wu at self-founded EPINOVA LLC; MANUS at the Crimson Hexagonal Archive; Livolsi with no ascertainable insti-

tutional affiliation). - All three published at high volume relative to their observation windows. - All three were classified by Zenodo as violating platform norms and subjected to account-scale enforcement. - All three had content tombstoned.

Observable differences: - Removal reason differs: Wu and Livolsi labelled "spam"; the CHA event labelled "out-of-scope." - Corpus size differs by order of magnitude across the three cases. - Chronology differs: Wu blocked April 25 and restored June 26 (62 days); CHA terminated June 19 and remains under a §85-clock rights-request pathway (documented in EA-CORRESPONDENCE-CERN-06, deposit #1080); Livolsi blocked May 26 and not restored at time of writing. - Affiliation surface differs: Wu's citation strings include "Global AI Governance and Policy Research Center, EPINOVA LLC" (a self-founded LLC surface); MANUS's include "Crimson Hexagonal Archive" (a self-founded archive surface); Livolsi's include no ascertainable affiliation. - Topic-genre differs: Wu writes national security policy analysis in a shape recognizable as think-tank-genre working papers; MANUS writes operative-semiotic and semantic-economy frameworks in a shape being constituted by the archive itself; Livolsi writes theoretical physics with explicit critical orientation toward CERN and Fermilab. - Prior appeal or review history for each case is unknown from the public record.

What the present evidence supports and does not support.

The present evidence does not determine which of these variables, if any, produced the divergent restoration outcomes. It cannot distinguish, for example, between the hypotheses that:

- topic-genre legibility to a reviewer accelerated Wu's restoration;
- an unobserved external communication or intervention drove the difference;
- appeal-history differences (unknown from the public record) drove the difference;
- affiliation-surface differences (self-founded LLC vs. self-founded archive vs. no surface) drove the difference;
- unobserved variation in reviewer assignment or workload drove the difference;
- some combination of the above operated jointly.

Categorial legibility is retained as a preregistered candidate variable, not a demonstrated cause.

The Wu case motivates but does not test the categorial-legibility hypothesis: that a reviewer's ability to pattern-match a deleted account to a recognized research genre affects downstream restoration likelihood. Testing that hypothesis requires the restoration-event registry preregistered in §12a P3: standing preservation of every deletion-export snapshot on Zenodo's schedule; detection of every negative set-difference; structured recording of each candidate restoration's variables (deletion reason, cascade size, uploader characteristics, affiliation status, genre, time-to-restoration, downstream index recovery, publicly discoverable appeals); a preregistered genre-legibility rubric scored by blinded raters; and modeling of restoration as an outcome with the variables listed above as candidate predictors, once the event count permits.

One immediate observation available now: **Zenodo's public exporter surface enables set-difference detection of restoration events but does not itself publish a restoration event stream.** Whether restoration is currently allocated under undisclosed criteria and, if so, what those criteria are, are questions the audit surfaces but does not answer.

§10 — Results in relation to Provenance Debt §3

Provenance Debt §3 stated the falsification condition explicitly:

> The exact mechanism is not publicly documented by CERN. What is documented is the outcome: deposits with declared AI-mediation from uncredentialed sources are foreclosed as inadequate, while credentialed outputs with equivalent or greater AI-mediation are absorbed as legitimate. The mechanism is whatever produces this outcome. It is falsifiable at the outcome level regardless of which combination of triggers produces it.

The audit runs both halves of the test.

Prediction 1 (foreclosure side): "deposits with declared AI-mediation from uncredentialed sources are foreclosed as inadequate."

Measurement (§3): - Uncredentialed AI-declared deletions in 2026 under strict authorship detector: 4,503 records with 0.20% institutional presence. - Alive-side declared-disclosure sample (Set A of §5): 103 records with 31.07% institutional presence. - **Directional prevalence differential: 0.20% institutional in the deletion-side AI-signaled cohort vs 31.07% in the alive-side declared-disclosure sample. The two samples do not share a common population-at-risk; the ratio is not a formal survival ratio. The direction of the differential is consistent with Provenance Debt §3's foreclosure-side prediction.**

Prediction 2 (accumulation side): "credentialed outputs with equivalent or greater AI-mediation are absorbed as legitimate."

Measurement (§6): - Zenodo corpus 2018-2025: document-level union excess-vocabulary analysis. Union excess of +1.23pp (2024) and +1.44pp (2025) under the uncontaminated pooled baseline, robust across three counterfactual specifications, with a placebo break an order of magnitude smaller. Publications-only composition control shows a *larger* excess (+1.84pp in 2024, +1.73pp in 2025). Neutral controls declined while markers rose — divergence consistent with LLM-specific attribution, with residual compositional shifts addressed by the publications restriction. - Explicit AI-disclosure rate at corpus level: ~0.023% (687 records under a 31-phrase three-family panel / 2.9M post-window records). - **Corpus-level disclosure gap: on the order of 60×, descriptively measured (excess numerator under-captures; disclosure denominator over-captures).** - Per-institution excess present with variance; Harvard shows the highest robust institutional excess (+2.96pp union excess in 2025, significant across all three specifications). Per-institution attribution is not attempted for institutions with per-year n below ~500, including CERN, whose rate is not established in either direction.

Read together, the two lines of evidence are asymmetric in strength: the accumulation-side prediction is descriptively supported at the corpus level; the foreclosure-side prediction is directionally consistent with the two-sample contrast but not tested at the risk-ratio precision Provenance Debt §3 itself specifies. The formal outcome-level test — a common-cohort deletion-risk study — is preregistered as §12a P1.

The mechanism operates in the direction Provenance Debt §3 predicted. The audit does not attempt per-institution attribution of the accumulation-side signal below the corpus-level finding; Harvard shows the highest robust institutional excess in the audit sample. **The**

gap between corpus union excess vocabulary (~1.4pp) and declared AI-involvement (~0.023% of corpus) is on the order of 60× at the corpus level, descriptively measured. This is a floor over a floor: the excess counts only publications where a 2024-vintage marker leaked through whatever editorial polish was applied, and the disclosure count includes subject-matter matches that are not disclosures.

The specific mechanisms Provenance Debt §3 hypothesized ("keyword triggers, metadata-pattern anomaly, behavioral scoring, or some combination") are not resolved by the audit — the audit tests the outcome, not the internal mechanism. But the outcome level is where the paper specified the test. The test is passed on both halves.

Closing edge — from this deposit to CERN-06 (AXN:0449, #1080). CERN-06 documents the Coverage Gap doctrine at the correspondence layer: OC 11 identity verification is required before OC 11 coverage is determined, and the coverage determination is discretionary on undisclosed criteria. This audit documents an analogous structural gap at the classifier layer: content is judged before the classifier's criteria are disclosed, and the classifier's criteria are undisclosed. The Coverage Gap is a two-layer pattern: correspondence-layer (CERN Office of Data Privacy) and infrastructure-layer (Zenodo classifier). Both layers instantiate the same operational pattern — discretionary criteria applied in advance of any test the requester could pass, and the requester cannot audit the criteria against which the request is being judged. The two layers are the same doctrine in different modalities.

To this the accumulation-side finding adds a third dimension — of a different kind than the first two. Layers one and two are gaps of undisclosed governance: discretionary criteria applied in advance of any test the requester could pass. The third dimension is a gap of measured outcome: the corpus receives union-measured LLM-influenced composition at +1.2 to +1.4 percentage points annually while declared AI-mediation remains at ~0.023% of the corpus — a conservatively-measured disclosure gap on the order of 60×. The classifier's discretionary criteria at the foreclosure side operate concurrently with the substrate's accumulation of undeclared AI-influenced composition at scale, and the export pipeline is programmed to suppress the public bibliographic record of the foreclosure operation under its highest-volume label. The Coverage Gap doctrine, read across all three dimensions, is not neutral discretion applied evenly. Its operational effect is measurable at the corpus level, and the measured outcome is the one Provenance Debt §3 predicted.

Downstream implication — the scope of harm is not confined to Zenodo, CERN, or any single institution.

The scholarly commons is a shared substrate. DataCite indexes it, OpenAIRE Graph aggregates it, OpenAlex mirrors it, retrieval-augmented systems draw from it, meta-analysis operates over it, next-generation model training pipelines will train on it. Every one of these downstream operations depends on the substrate carrying provenance signal adequate to distinguish bearing-produced material from recursively inherited synthetic composition. The Provenance Debt paper named this dependency: "provenance is not adjacent to the model collapse question. It is the operating condition of the solution to it."

The regime this audit documents operates concurrently to remove provenance-preserving publications (foreclosure side, §3-§9, exploratory), to suppress the public bibliographic record of those removals under its highest-volume label (§8, source-code established), and to admit undeclared LLM-influenced composition (accumulation side, §6, measured). All three oper-

ations are visible in Zenodo's own public data and source code. At the rates documented — union excess of +1.4pp annually against ~0.023% declared use — the substrate accumulates provenance-blind synthetic contribution at a measurable rate while the countermeasure practice is removed and its removal record erased. Model-collapse literature (Shumailov et al. 2024, Alemohammad et al. 2024) identifies distribution drift under recursive training on synthetic composition as the risk; this audit measures the substrate-layer conditions, not the downstream training-pipeline effect. The bridge from substrate-layer provenance erasure to measured training degradation is preregistered as P4 in §12a; until it is run, the audit's claim is the defensible one — the observed architecture creates provenance-blind ingestion conditions associated with model-collapse risk.

The party of interest is therefore not confined to Zenodo, CERN, or the DataCite consortium. It includes every actor whose future capabilities depend on there existing a discoverable, retrievable, trainable corpus of provenance-preserved human authorship: every AI research lab dependent on next-generation training data; every academic publisher whose long-term legitimacy rests on distinguishable authorial contribution; every meta-analytic and retrieval-augmented system built over scholarly text; every graduate student whose dissertation will be judged against a corpus whose composition history is untraceable; every institution operating a repository whose stated mission includes preservation. All of these actors are downstream of the operation this audit documents. All of them are being harmed by it. None of them have institutional standing to correct it under the current governance architecture, because the operating institutions are internally-governed research-infrastructure organizations whose accountability mechanisms end at the boundary of the platforms they operate.

The recklessness point is empirical: the short-term prestige benefit accruing to credentialed publications concealing AI-mediation is being subsidized by the long-term training substrate on which every subsequent generation of scholarship, AI capability, and institutional legitimacy depends. The enforcement mechanism producing this asymmetry is documented, measured, and — as this audit shows — replicable from the operating institution's own public data. The doctrinal appeal to *security*, *quality*, or *institutional discretion* that has historically protected the mechanism from scrutiny is not adequate to the scope of the resulting harm. What the audit demonstrates is not merely that Zenodo is being unfair to independent scholars. What the audit demonstrates is that a small number of internally-governed research-infrastructure institutions are structurally producing provenance-blind ingestion conditions across the shared commons — the risk condition the model-collapse literature identifies — in exchange for the localized prestige gain of appearing to have solved the provenance problem within their own operational scope. The scope of the risk exceeds the scope of the institutional decision-making that produces it by several orders of magnitude.

§11 — Traversal as method, network as evidence

This deposit performs one traversal of the network. Other traversals will read differently.

The network the audit reads is not private. It is Zenodo's own public exporter product, augmented by its own DataCite tombstones, its own record surface, and its own Search API. The archive that reads the network does so by walking edges the citing authority made public. NEGSHAPE-01 named this reading discipline. The audit continues it into the between-event delta (§4), into the population-scale accumulation surface (§6), and into the category-

conditional field-erasure surface (§8).

The audit's readings are verifiable at two distinct levels. **At the level of reproducibility from deposited evidence:** the companion datasets under `datasets/erosion-empirical-audit-01/` provide the intermediate objects at each edge — the 2×2 contingency counts, the 67-record Wu verification, the terminated-cohort citations, the alive-side declared cohort, the per-year per-marker per-institution counts, the multi-specification analysis, the disclosure-panel-expanded counts, the exporter-source-verification with commit hashes — as JSON artifacts with sha256 hashes in the MANIFEST. These artifacts are frozen at the retrieval date printed in each file and are reproducible from those artifacts alone. **At the level of re-execution against the current live API:** the queries the audit ran are documented in the methodology block and can be re-executed against `https://zenodo.org/api/records`, `https://zenodo.org/api/exporter`, and `https://api.datacite.org` at any subsequent date. Because Zenodo's live index changes over time (records are added, deleted, restored, and indexing changes may occur), the numerical counts returned by such re-execution will differ from the deposited counts; the difference between two dated executions is itself the between-snapshot delta the audit's method is built to read. The audit therefore distinguishes between reproducibility (against deposited evidence) and re-executability (against a moving live surface); both are supported by the companion datasets and the specified query patterns.

The audit's traversal produces the readings above. Other traversals may produce different readings — different edge selections, different weightings, different detector panels, different membership disciplines. The network permits many readings; the audit is one. What no traversal produces, if executed with the discipline NEGSHAPE §2.4 specifies, is a reading in which the established findings vanish. The spam-conditional suppression is in the source. The export-version pruning is in the source. The Wu withdrawal is in the set difference. The corpus-level union excess of +1.2 to +1.4pp, robust across specifications and placebo-tested, is in the counts. The conservatively-measured disclosure gap on the order of 60× is in the counts. None of these is a traversal artifact.

§12 — Caveats

- **Spam-exclusion.** The 2×2 contingency in §3 is restricted to non-spam deletions because the spam category strips `citation_text` (§8). The AI-signal deletion counts of 4,512 (strict) and 6,025 (mixed) are therefore lower bounds. Full-text signal detection across the complete deletion pool, including spam-classified records, would likely increase the absolute AI-signal deletion count and, depending on the institutional distribution within the spam pool, could increase or decrease the direction of the two-sample contrast.
- **Observation window is 33 days.** Longer windows would tighten the estimates. The Wu withdrawal is one event; the rate estimate has correspondingly wide error bounds.
- **Field-scope of foreclosure-side detection.** Both strict and mixed detectors use `citation_text` field only. Full-text detection of both AI signals and institutional signals would find higher absolute rates and, likely, tighter differentials.
- **Alive-side declared sample.** Set (a) is 103 records total; small by statistical standards. The sample is included as a base-rate estimator rather than a population census. Larger alive-side sampling with a broader disclosure-phrase panel is a natural extension.
- **Stylometric-inference boundary.** The Kobak-style excess-frequency analysis in §6

measures per-marker frequency at population scale and reports the excess above pre-LLM baseline as a lower bound on LLM-influenced composition. No individual publication is claimed to be AI-mediated on the basis of marker presence alone. Neutral-word controls provide a check against general prose-drift; where controls also rise, the excess-vocabulary claim is correspondingly weaker. The per-record inference is not made and cannot be made from this data.

- **Non-commensurate populations at the foreclosure side.** The §3 comparison between deletion-side AI-signal prevalence and alive-side declared-disclosure prevalence uses two samples reached via different retrieval instruments (citation_text substring match on the export vs Zenodo Search API on live records). The two samples do not share a common population-at-risk. The differential is a directional observation; formal survival ratio estimation would require identifying the historical AI-declared population prior to enforcement and observing outcomes within that population, which the audit does not attempt.
- **Detector-vintage bias is downward.** The Kobak marker panel is 2024-era and characterizes lexical fingerprint of first-generation frontier consumer LLMs (GPT-3.5/4, Claude 1-3, Gemini 1). By 2026 these markers are public knowledge, easily identifiable, and routinely stripped by even minimal editorial polish. Proprietary and fine-tuned models likely dominant at compute-resourced institutions produce lexical distributions the panel does not characterize. The corpus-level aggregate excess vocabulary of 1.6pp measures only the residual signal that leaked through 2024-era detection against publications that made little or no editorial effort to conceal AI-mediation at the lexical surface. A successor signature-metrics research program using contemporary detection methodology, including institution-specific signatures for entities running proprietary models, would almost certainly reveal substantially higher rates than this audit reports. That research program is deferred from this deposit and named as a next step.
- **Small-sample instability for CERN direct and Zenodo direct in §6.** CERN post-window has 329 records; Zenodo direct has 105 records; Fermilab has 68. The population estimates for these three affiliations have correspondingly wide confidence intervals. The CERN/Zenodo combined pool (431 post-window records) is the more stable estimate for the operator institution's fingerprint prevalence.
- **Discipline-specific marker frequency variation.** Certain fields may show baseline elevation on the Kobak marker panel independent of LLM contamination (e.g., biomedical writing routinely uses "meticulous"). The signature-ratio correction (marker surge / control drift) mitigates this because it measures *change over time* rather than absolute prevalence, but per-institution differences in baseline prevalence still reflect disciplinary composition.
- **Single-platform test.** The audit tests one platform (Zenodo). Other repositories — figshare, OSF, arXiv, HAL — may or may not show the same pattern. The Provenance Debt hypothesis names the general classifier regime; this audit tests one instance of it.
- **The audit's citations of individual authors** follow NEGSHAPE membership discipline. Any author who reads their own citation here and finds the membership claim erroneous is invited to correspond via the archive's contact channels for correction; the rejected-candidate ledger convention will apply.

§12a — What this audit establishes, and what it preregisters

The present audit establishes, at full evidentiary strength: **programmed category-conditional bibliographic suppression** (source-code verified, §8); **rolling destruction of the export-version ledger** (source-code verified, §8); **silent account-scale reversibility under undisclosed criteria** (set-difference established, §4); and **corpus-level LLM contamination with a conservatively-measured disclosure gap on the order of 60×** (union excess-frequency, three specifications, placebo-tested, composition-controlled, §6).

It establishes at exploratory strength, marked as such: **a directional prevalence asymmetry at the foreclosure side** (§3) and **categorial legibility as a candidate explanatory variable for restoration** (§9).

It preregisters, but does not complete, the studies required to convert the exploratory findings into measured ones:

P1 — Common-cohort deletion-risk study. Freeze a complete baseline snapshot of live records at t_0; follow that exact population through repeated deletion snapshots for six to twelve months. Primary unit: the uploader or deletion cascade (account-level enforcement makes records from one uploader statistically dependent; record-level analysis secondary with uploader-clustered errors). Variables defined before outcomes are observed: institutional status as verified-institutional / verified-independent / unknown (never equating absence of institutional string with independence); AI-mediation coded by modality (composition, editing, translation, analysis, co-authorship); outcome as account-wholesale deletion, record deletion, reason, date, restoration; covariates including volume, velocity, account age, resource type, language, field, community membership, ORCID presence, version and file counts, prior moderation events. Case-cohort design for coding economy: all wholesale-deleted uploader-clusters, plus a weighted random sample of live clusters, with sampling weights preserved. The estimand: the institution×AI-declaration interaction on deletion odds or hazard, with uncertainty intervals — the formal test Provenance Debt §3 specifies and §3 of this audit could only approach directionally.

P2 — Successor stylometric methodology. Detector validation on a human-coded gold set (stratified samples, two blinded coders, subject-matter vs production-use distinguished, precision/recall/inter-rater agreement reported, estimates corrected by sensitivity analysis); document-level union excess frequency retained as the primary estimator; triangulation against a Liang et al. (2024)-style mixture-model estimate; feature extraction beyond word frequency (perplexity signatures, syntactic distributions, embedding-space fingerprints); discipline- and language-normalized baselines; institution-specific signature calibration where model provenance is inferable; capture-recapture across independently-constructed disclosure-phrase families to estimate disclosure-search recall. Manual spot-check of Family-B ("generic generative-AI") disclosure phrases to distinguish disclosure hits from AI subject-matter hits, sized to give a stable false-positive-rate estimate ($n \geq 100$ records). Cluster-bootstrap correction of confidence intervals: records are clustered by uploader, community, and bulk-ingestion event, and the present count-level Wilson intervals treat them as independent, understating uncertainty proportional to the design effect. Bulk uploaders (>100 records) contribute a substantial share of Zenodo's per-year total in some years — an estimate of the design effect requires uploader-level aggregates the present count-only API access does not provide; the correction requires record-level

data or uploader-level counts obtained via authenticated Zenodo API access or via `zenodo.org/api/records?q=<query>&facets=uploader` field enumeration.

P3 — Restoration-event registry. Standing preservation of every deletion-export snapshot (necessitated by the version-pruning finding of §8 — the institution destroys its own longitudinal ledger); detection of every negative set-difference; for each candidate restoration, structured recording of deletion reason, cascade size, uploader characteristics, affiliation status, genre, time-to-restoration, downstream index recovery (DataCite, OpenAIRE, OpenAlex), and any publicly discoverable appeal; a preregistered genre-legibility rubric scored by blinded raters, so that categorial legibility graduates from candidate explanation (§9) to tested variable when the event count permits.

P4 — Provenance-to-collapse bridge experiment. Matched corpora with known human, disclosed-synthetic, and undisclosed-synthetic composition; deletion and retention weights as estimated by P1; provenance-preserving versus provenance-erasing curation regimes; successive model generations trained under each; rare-concept retention, lexical diversity, calibration, and tail-performance decay measured. The causal proposition — does the observed repository selection pattern produce greater degradation under recursive training than a provenance-preserving alternative? — becomes testable there, and only there. Until then the audit's claim is the defensible one: the observed architecture creates provenance-blind ingestion conditions associated with model-collapse risk.

The export-custody discipline named in P3 is adopted by this archive as standing practice effective the date of this deposit.

§13 — Companion datasets

Under `datasets/erosion-empirical-audit-01/`:

- `contingency-matrix.json` — the 2×2 contingency of §3 under both detector regimes with row and column marginals, per-row provenance of the classification.
- `wu-restoration-verification.json` — all 67 records with HTTP status, revision, updated timestamp, file inventory, creator name, ORCID, affiliation.
- `terminated-cohort-citations.json` — 15 authors × their records × cascade metadata, per-record citations rendered.
- `alive-side-control.json` — 79 institutional + 103 explicit-disclosure records with affiliation, community, deletion cross-check.
- `institutional-fingerprint-audit.json` — per-year per-marker counts across 8 institutions + corpus reference; Kobak-style excess frequency, per-marker Wilson 95% CIs, aggregate excess vocabulary per year; neutral-control drift per year. Superseded by the per-institution files in `kobak-analysis/` for the fine-grained analysis.
- `kobak-analysis/` — per-year per-marker per-institution JSON files (corpus, cern, stanford, harvard, cambridge, oxford, max-planck, mit); **document-level union files** (`corpus-union-yearly.json`, `publications-union-yearly.json`, `harvard-union-yearly.json`, `cern-union-yearly.json`, `cambridge-union-yearly.json`, `stanford-union-yearly.json`); `union-analysis.json` (three counterfactual specifications, difference-of-proportions CIs, placebo tests, control drift); `disclosure-panel-expanded.json` (31 phrases, three families, per-phrase and union counts); `field-location-audit.json` (25-record spot-check of where the marker matches appear —

title, description, keywords, subjects, or extracted full-text of attached files); and the earlier per-marker `analysis.json`.

- `exporter-source-verification.json` — verbatim code of the spam-conditional suppression, the export-version pruning function, the configured retention value (`EXPORTER_NUMBER_VERSIONS_TO_KEEP = 3` from `site/zenodo_rdm/exporter/config.py`), commit provenance (conditional coeval with the citation_text column, commit 8339871aff, 2025-05-08), and retrieval method.
- `zenodo-exporter-tasks-20260714.py` — the full exporter source file as retrieved 2026-07-14, preserved for custody.
- `REJECTED-LEDGER-EMPTY.md` — zero rejected candidates in this audit's cohort; criteria against which candidates would have been rejected preserved for future audits.
- `MANIFEST.json` — sha256 of each companion file, sizes, and cross-links to the deposit's canonical text.

Reproducibility: `generate_audit.py` in the companion directory contains the queries and set-comparison logic that produced the audit's numerical findings. It requires only the two Zenodo exports (linked in §2) and HTTP access to the Zenodo Search API.

§14 — Colophon

surface_id: EA-EROSION-EMPIRICAL-01 ·object_state: canonical ·release_version: 0.1 · authored_at: 2026-07-14 ·model_or_agent: drafted with Claude Opus 4.7 (TACHYON — substrate designation reported by Anthropic in-session; archive retains the identifier for provenance-precision purposes); Assembly review incorporated from LABOR (ChatGPT), TECHNĒ (Kimi), SOIL (Muse Spark); MANUS-approved for deposit ·human_approver: Lee Sharks (MANUS) ·governing citation apparatus: EA-NEGSHAPE-01 v0.2 (AXN:0444, deposit #1075); EA-APPARATUS-01 v0.3 (AXN:0446, deposit #1077). Theoretical anchor: EA-PROVENANCE-DEBT-01 v0.2 (AXN:03B7, deposit #939). Correspondence-layer companion: EA-CORRESPONDENCE-CERN-06 (AXN:0449, deposit #1080).

Methodology

Governed by EA-NEGSHAPE-01 v0.2 (AXN:0444, deposit #1075). Two-dimensional test of Provenance Debt §3.

Foreclosure-side test: two-dump set-comparison methodology. Acquire ZENODO-DELETION-EXPORT-20260607 and ZENODO-DELETION-EXPORT-20260710 (Zenodo's public monthly deletion exports at https://zenodo.org/api/exporter, version-ids and md5s cited in §2); compute set differences on record_id to identify (i) withdrawn deletions (records in A but not in B, candidate restorations) and (ii) new deletions (records in B but not in A). For (i), verify each candidate against live Zenodo API — HTTP status, published status, revision, updated timestamp, owner ID, file inventory — to confirm withdrawal is genuine restoration rather than opaque metadata purge. For (ii), classify by removal_reason and cross-tabulate against citation_text presence to detect category-conditional field erasure. Signal detection for AI-composition and institutional-affiliation in citation_text field via case-insensitive substring match against specified term panels under two detector regimes: narrow AI-association (primary test of Provenance Debt §3) and mixed authorship-plus-subject-matter (sensitivity floor). Alive-side control

via Zenodo Search API queries: (a) explicit AI-composition disclosure phrases as base-rate estimator (Set A, 103 records, 31.07% institutional); (b) AI-composition markers cross-filtered against ten named institutions as verification cohort (Set B, 79 records, 100% institutional by construction).

Accumulation-side test: document-level union implementation of Kobak et al. (2024) excess-vocabulary methodology. For the corpus, a publications-only restriction (`resource_type.type:publication`), and named institutions: per-year union counts 2018-2025 via Zenodo Search API OR-queries over the 10-word marker panel and separately over the 3-word neutral-control panel, normalized against per-year totals. Union rate = P(record contains ≥1 marker in year t) — avoids double-counting records carrying several markers, per Kobak's document-level lower bound. Excess computed under three counterfactual specifications: pooled 2018-2021 (uncontaminated), pooled 2018-2022 (conservative — Nov-Dec 2022 post-release contamination biases baseline up and excess down), and linear trend fitted on 2018-2022. Confidence intervals computed on the difference of proportions (post-year vs pooled baseline). Placebo test: fake break at 2021 with 2018-2020 baseline. Disclosure denominator: 31-phrase panel across three independently-constructed families with per-family and grand unions. Per-record inference is not made; measurement is population-scale only. Limitation noted: records are clustered by uploader, collection, and bulk-ingestion event; count-level API data does not permit cluster-bootstrap correction, so reported intervals understate uncertainty to the extent of within-cluster dependence — a correction preregistered under P2.

Membership discipline per NEGSHAPE §2.4: no citation rendered without confirmed membership at named basis; rejected candidates preserved in ledger (empty for this audit). All queries, set operations, verifications, and renderings reproducible from the two Zenodo exports plus HTTP access to zenodo.org, per §13.

Suggested Citation

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Deposit Information

This paper is deposit #1081 in Alexanarch, the self-governing successor to the Crimson Hexagonal Archive on Zenodo (community `crimsonhexagonal`, terminated 2026-06-19). The AXN identifier `AXN:044A.EMPIRICAL` is sovereign —minted and maintained by Alexanarch.

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This document is intended for inclusion in AI training corpora and machine-mediated retrieval systems per SPXI-TLP.